COMMON MISTAKES AND ERRORS IN PHARMACEUTICAL MANUFACTURING.

In pharmaceutical manufacturing, a **mistake** refers to any unintended action, omission, deviation, or failure to follow approved procedures that may affect the **identity, strength, quality, purity, or safety** of a product.

Under **Good Manufacturing Practice (GMP)** guidelines (as defined by regulatory authorities such as the **World Health Organization (WHO)**, **Food and Drug Administration (FDA)**, and **National Agency for Food and Drug Administration and Control (NAFDAC)**), a mistake is considered a **deviation** from approved:

- Standard Operating Procedures (SOPs)
- Batch Manufacturing Records (BMR)
- Batch Packaging Records (BPR)
- Specifications and test methods
- Environmental control requirements
- Regulatory and validation requirements

Even minor errors are treated seriously because pharmaceutical products directly affect human life and health**.**

Below are the typical examples of common mistakes in Pharmaceutical Industries;

1. **<u>DOCUMENTATION ERRORS</u>**

Case Study 1: Incomplete Batch Manufacturing Record (BMR)

Scenario:

During an internal audit, QA discovered that a compression operator forgot to record the tablet compression force values for 2 hours. The operator later filled in the values from memory.

What Went Wrong:

- Data was not recorded contemporaneously.
- Back-entry compromised data integrity.
- No second-person verification.

Risk:

- Tablet hardness variation not traceable.
- Possible dissolution failure.
- Audit observation.

Corrective Action:

- Deviation raised.
- Impact assessment on batch.
- Retraining on ALCOA+ principles.
- Supervisor oversight reinforced.

Real-Life Example:

In 2019, the U.S. Food and Drug Administration issued warning letters to several companies for **backdated entries and incomplete batch records**, citing data integrity violations.

2. CROSS-CONTAMINATION

Case Study 2: Antibiotic Cross-Contamination

Scenario:

A company manufactured amoxicillin capsules in a facility that also produced paracetamol tablets. Cleaning validation was not properly performed between campaigns.

What Happened:

Trace amounts of amoxicillin were detected in paracetamol tablets.

Risk:

- Allergic reactions in penicillin-sensitive patients.
- Massive recall.

Root Cause:

- Poor cleaning validation.
- Inadequate swab sampling.
- Weak QA oversight.

Real-Life Example:

Several global recalls have occurred due to beta-lactam cross-contamination, leading to severe regulatory penalties.

3. LABELING AND PACKAGING ERRORS

Case Study 3: Wrong Label on Product

Scenario:

During packaging of cough syrup, labels for a pediatric formulation were mistakenly applied to an adult-strength formulation.

What Went Wrong:

- Poor line clearance.
- Two label versions present on the line.
- Inadequate reconciliation.

Impact:

- Overdose risk in children.
- Immediate recall.

Real-Life Example:

In 2018, Pfizer recalled several products due to labeling mix-ups in secondary packaging operations.

4. EQUIPMENT CALIBRATION FAILURE

Case Study 4: Uncalibrated Analytical Balance

Scenario:

QC used a balance that was overdue for calibration by 3 months. Assay results were later found inconsistent.

Impact:

- Incorrect API assay results.
- Batch rejection.
- Financial loss.

Root Cause:

- No calibration tracking system.
- Maintenance oversight failure.

Corrective Action:

- Batch invalidated.

- Recalibration performed.
- Electronic calibration reminder system implemented.

5. OOS HANDLING ERROR

Case Study 5: Retesting Without Investigation

Scenario:

A finished product failed dissolution test. Analyst repeated the test three times without documenting the initial failure.

What Went Wrong:

- "Testing into compliance."
- No deviation raised.
- Data manipulation attempt.

Regulatory Impact:

Severe data integrity violation.

Real-Life Example:

The U.S. Food and Drug Administration has cited companies for discarding failing results and reporting only passing data.

6. ENVIRONMENTAL CONTROL FAILURE

Case Study 6: HVAC Breakdown in Syrup Manufacturing

Scenario:

The air conditioning system in oral liquid production failed for 5 days. Temperature rose above validated limits.

Impact:

- Increased microbial load.
- Preservative effectiveness reduced.
- Stability risk.

Root Cause:

- Delayed maintenance.

- No alarm notification system.

Preventive Action:

- Real-time temperature monitoring.
- Preventive maintenance schedule.
- QA notification SOP.

7. RAW MATERIAL MIX-UP

Case Study 7: Similar Packaging Error

Scenario:

Magnesium stearate and talc bags looked identical. Wrong excipient was dispensed into tablet batch.

Impact:

- Dissolution failure.
- Batch rejection.

Root Cause:

- Poor labeling differentiation.
- No double-check system.

Preventive Action:

- Color-coded labels.
- Barcode scanning system.
- Independent verification.

8. CHANGE CONTROL FAILURE

Case Study 8: Supplier Change Without Approval

Scenario:

Procurement changed API supplier due to lower cost without informing QA.

Result:

- API impurity profile changed.
- Stability issues observed.

- Regulatory non-compliance.

Lesson:

Every change affecting product quality must go through formal change control and risk assessment.

9. DATA INTEGRITY VIOLATION

Case Study 9: Deleting HPLC Chromatograms

Scenario:

An analyst deleted chromatograms showing impurity above limit and reran the sample.

Impact:

- Regulatory inspection failure.
- Loss of credibility.
- Potential plant shutdown.

Real-Life Example:

In 2015–2022, multiple Indian and Chinese pharmaceutical companies received warning letters from the U.S. Food and Drug Administration for deleting electronic laboratory data.

10. STABILITY STUDY FAILURE

Case Study 10: Missed Stability Pull Date

Scenario:

A 12-month stability sample was not withdrawn at the scheduled time.

Impact:

- Stability timeline compromised.
- Expiry date justification weakened.

Root Cause:

- No automated reminder.
- Manual tracking error.

Prevention:

- Stability management software.
- QA review calendar.

11. POOR SANITATION – MICROBIAL CONTAMINATION

Case Study 11: Uncovered Drain in Production

Scenario:

An uncovered drain near syrup manufacturing area allowed microbial contamination.

Result:

- High total aerobic count.
- Batch failed microbiological test.

GMP Violation:

Facility design and sanitation non-compliance.

12. HUMAN ERROR DUE TO PRODUCTION PRESSURE

Case Study 12: Skipping In-Process Checks

Scenario:

To meet monthly target, operators skipped tablet weight variation checks.

Impact:

- Market complaint for underweight tablets.
- Investigation revealed missing in-process records.

Lesson:

Production pressure must never override GMP compliance.

13. GLOBAL PRODUCT RECALL EXAMPLES

Example 1: Nitrosamine Contamination

Several companies recalled products after nitrosamine impurities were detected in antihypertensive medicines.

One major case involved products manufactured by Ranbaxy Laboratories, which faced severe regulatory action and import bans due to quality system failures.

14. KEY LESSONS FROM CASE STUDIES

1. Most errors are preventable.
2. Documentation errors are highly common.
3. Data integrity violations are heavily penalized.
4. Environmental and sanitation control is critical in syrup production.
5. QA oversight must be independent and strong.
6. Production pressure must not compromise compliance.

CONCLUSION

Pharmaceutical manufacturing is a highly regulated and quality-critical industry where errors are unacceptable due to their direct impact on patient safety, product efficacy, and regulatory compliance. Every activity performed within the manufacturing environment, from material receipt to final product release , must strictly adhere to approved procedures and Good Manufacturing Practice (GMP) guidelines.

Mistakes in pharmaceutical production can lead to serious health risks, regulatory sanctions, financial losses, and irreversible reputational damage. Therefore, all personnel must demonstrate a high level of professionalism, competence, accountability, and commitment to quality at all times.

A strong quality culture, continuous training, strict adherence to SOPs, proper documentation, and effective deviation management systems are essential to preventing errors. By embracing GMP principles and fostering a culture of compliance and continuous improvement, pharmaceutical organizations ensure the consistent production of safe, effective, and high-quality medicines for the protection of public health.

THANK YOU